



University of Idaho

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CS 4727/5727
Adversarial
Machine Learning

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Lecture 13

Bias and Fairness in Machine Learning



Lecture Content

13.1 Introduction to Bias and Fairness in ML

13.2 Introduction to ML Bias

13.2.1 Data Bias

13.2.2 Algorithmic Bias

13.3 Bias Amplification

13.4 Case Study: COMPAS

13.5 Fairness Metrics

13.5.1 Causal Fairness and Counterfactual Methods

13.6 Bias and Fairness in Large Language Models

13.7 Mitigation strategies

13.8 Governance, Legal Frameworks, and Ethical AI

Presentation by Kyle Russel Delos Santos

13.9 Dai (2024) Bias and Unfairness in Information Retrieval Systems: New Challenges in the LLM Era



Introduction

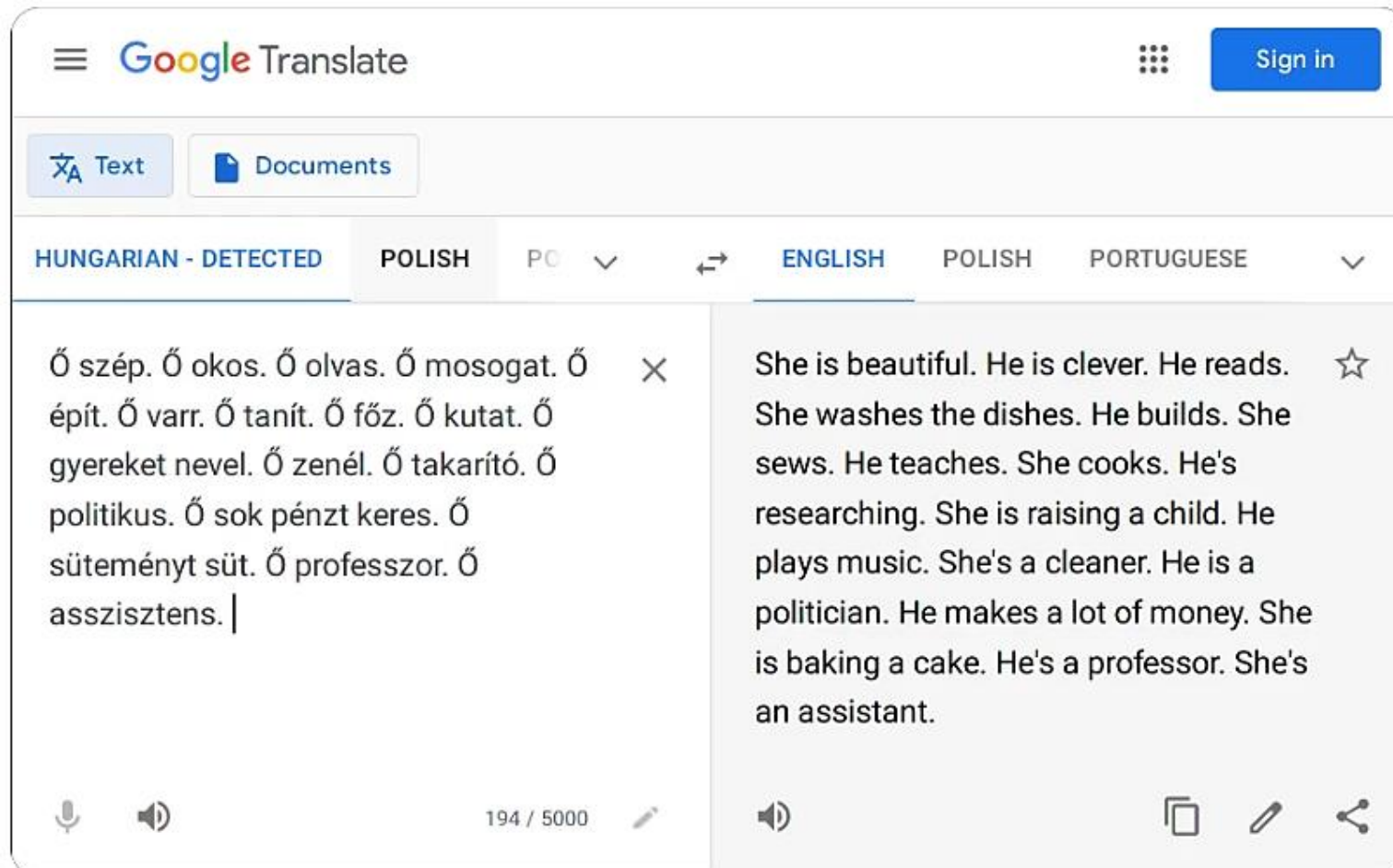
13.1 Introduction to Bias and Fairness in ML

- Although ML models are increasingly used to influence decisions that impact society, the models are not inherently neutral
- ML models reflect the biases present in their training data, potentially expanding and amplifying existing societal inequalities
- When deployed at scale, biased ML models can systematically disadvantage marginalized groups
 - A biased ML-based system that decides how to allocate resources may allocate resources unfairly or perpetuate inequality
- Examples:
 - Loan applications
 - Mortgage applications
 - Insurance rates
 - College admissions
 - Job applications
 - Sentencing criminals

Bias Example

13.1 Introduction to Bias and Fairness in ML

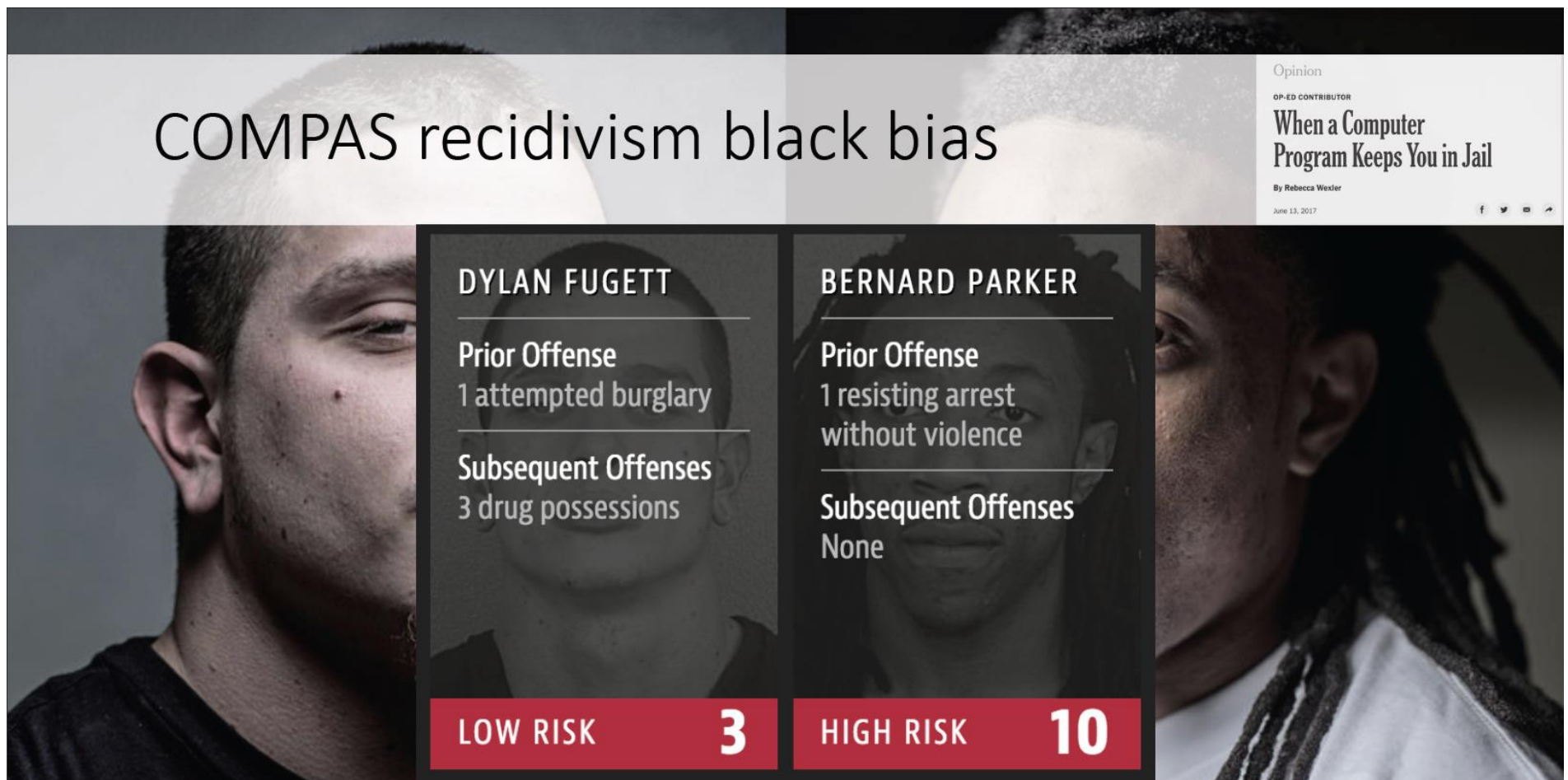
- Output from Google Translate for Hungarian to English translation
 - Hungarian has no gendered pronouns, Google Translate makes assumptions



Bias Example

13.1 Introduction to Bias and Fairness in ML

- An ML system named COMPAS produces biased results for crime offenders (the study is described later in this lecture in more detail)



Bias and Fairness Definitions

13.1 Introduction to Bias and Fairness in ML

- *Bias in ML and AI*
 - **Bias** is a systematic error in the ML model that produces skewed results due to assumptions, training data, or learning process
 - When the bias leads to unequal outcomes across demographic groups, it may result in unfairness
 - Note that an ML model can be biased in a statistical sense but still treat groups equally
- *Fairness in ML and AI*
 - **Fairness** is the absence of prejudice or favoritism toward individuals or groups based on their protected characteristics
 - Fairness extends beyond equality of treatment, providing different levels of support to achieve just outcomes across diverse populations
 - Multiple mathematical definitions of fairness exist, each representing different perspectives on what constitutes "fair" outcomes
- Understanding bias and fairness in ML requires recognizing that these concepts are nuanced and context-dependent
 - What constitutes "fair" in one application may not be appropriate in another application: this requires ongoing evaluation and adjustment



Introduction to ML Bias

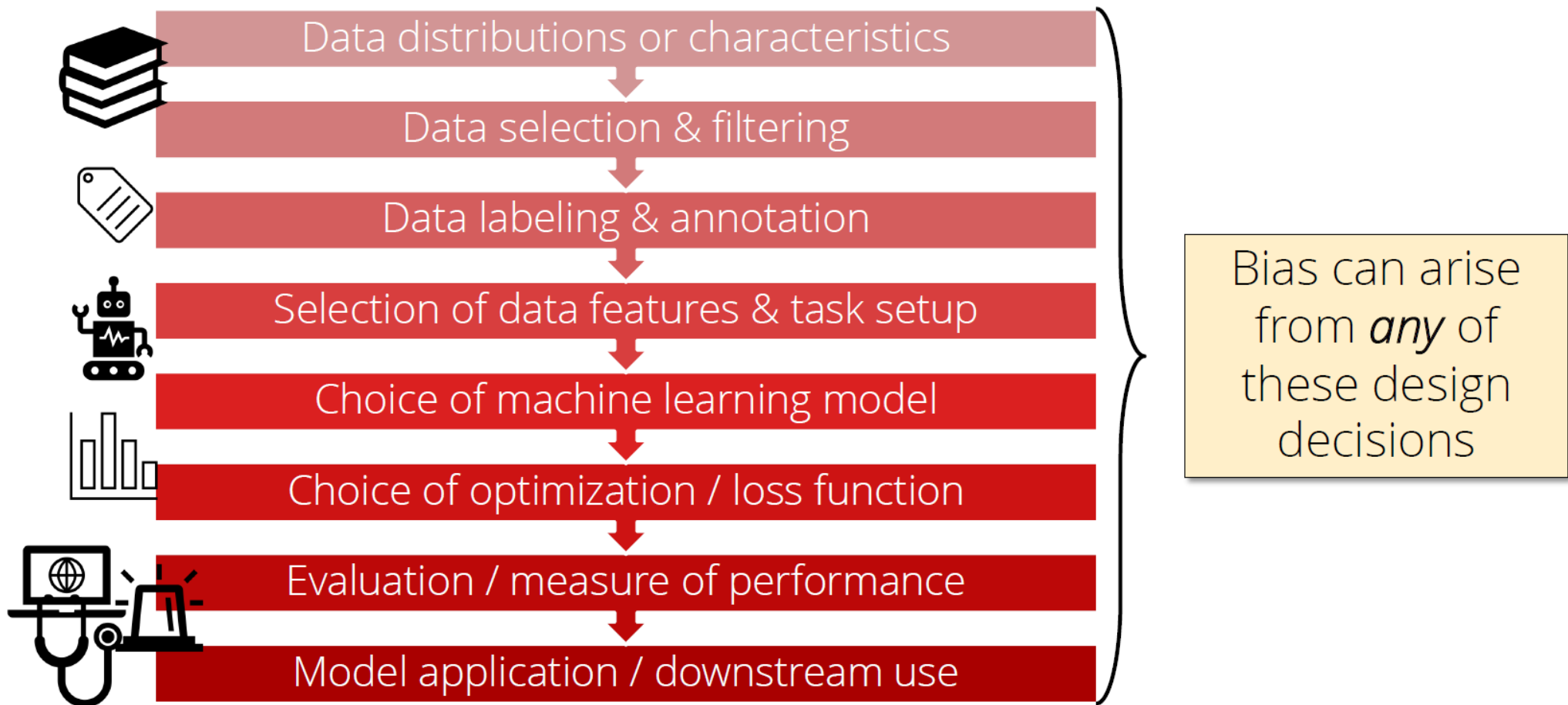
13.2 Introduction to ML Bias

- Sources of ML bias
 - **Data bias**
 - Data bias is one of the main reasons for fairness issues in ML systems
 - Recognizing the sources of data bias is important for building trustworthy AI, as ML models can only be as fair as the data used to train them
 - **Algorithmic bias**
 - Even with balanced training data, the choice of ML algorithm can introduce or intensify unfairness in model outcomes
 - Algorithmic bias can emerge at various stages of the ML pipeline, during model design, optimization, evaluation, and deployment
 - Assumptions in the ML model creation can amplify certain characteristics over others, leading to biased outputs

Sources of Bias in ML

13.2 Introduction to ML Bias

- All phases in the ML pipeline can introduce bias, from data preprocessing to model development and selection to downstream use



Types of Data Bias in ML

13.2.1 Data Bias

- **Historical bias**
 - Includes pre-existing historical societal inequalities reflected in data, even when the data is collected accurately
 - E.g., criminal justice data overrepresenting certain demographics
 - E.g., medical research data historically focused on male subjects
- **Data aggregation bias**
 - Combining data in ways that ignore disproportions between subgroups
 - E.g., averaging outcomes across different demographic groups
- **Representation bias**
 - Samples that fail to reflect the diversity of the population
 - E.g., facial recognition ML model trained primarily on light-skinned faces
 - E.g., voice recognition ML model struggling with certain accents
- **Omitted variable bias**
 - Arises when important features are excluded from the ML model
 - E.g., a credit scoring algorithm that considers only traditional credit history may disadvantage communities with limited access to banking services
- **Measurement bias**
 - Errors or inconsistencies in data collection or labeling
 - E.g., health sensors less accurate for darker skin tones
 - E.g., inconsistent labeling standards across different groups

Types of Algorithmic Bias in ML

13.2.2 Algorithmic Bias

- **Algorithm bias**
 - Originates from inherent limitations or design flaws in ML algorithms
 - Some ML model architectures may be more likely to amplify bias present in the training data, or may prioritize optimization methods or metrics that don't align with fairness considerations
- **Regularization bias**
 - Regularization techniques can suppress minority group information and treat that information as noise
 - E.g., L_1/L_2 regularization might eliminate rare but important features
- **Transfer bias**
 - Occurs when models trained in one context (domain, population) are reused elsewhere without proper adaptation
 - E.g., a healthcare model trained on U.S. patient data performs poorly on data from another country
- **Evaluation bias**
 - Occurs when the model is evaluated on a non-representative dataset
 - E.g., facial recognition evaluated on one demographic group leads to decreased performance when the system is deployed broadly across diverse populations



Algorithmic Bias in ML Systems

13.2.2 Algorithmic Bias

- [Safiya Umoja Noble \(2018\) Algorithms of Oppression](#)
 - Examines how search engines often prioritize content that reinforces racial stereotypes, showing how algorithmic structures reflect societal biases
- [Bender \(2021\) On the Dangers of Stochastic Parrots](#)
 - Discusses risks in LLMs emphasizing that over-reliance on biased data leads to “parroting” harmful or inaccurate stereotypes
 - Highlights the responsibility of tech companies in managing and reducing bias in generative AI systems
- [Humble \(2024\) War, Artificial Intelligence, and the Future of Conflict](#)
 - Explores ethical concerns around AI deployment in military and conflict scenarios
 - Raises awareness of how biased AI in conflict situations could disproportionately target specific groups

Bias Amplification

13.3 Bias Amplification

- One of the most concerning aspects of bias in ML is the potential for *bias amplification* effects
 - When ML models trained on biased data are deployed in real-world scenarios, they can create feedback loops that reinforce existing inequalities
- Example
 - ML algorithms direct more officers to neighborhoods with historically higher arrest rates
 - The increased police presence leads to more arrests
 - This generates more data points that reinforce the model's prediction that these areas have higher crime rates, regardless of the actual crime rate
- Similar feedback effects can occur in hiring, lending, healthcare, and other domains, creating a cycle that becomes difficult to break

Bias Amplification

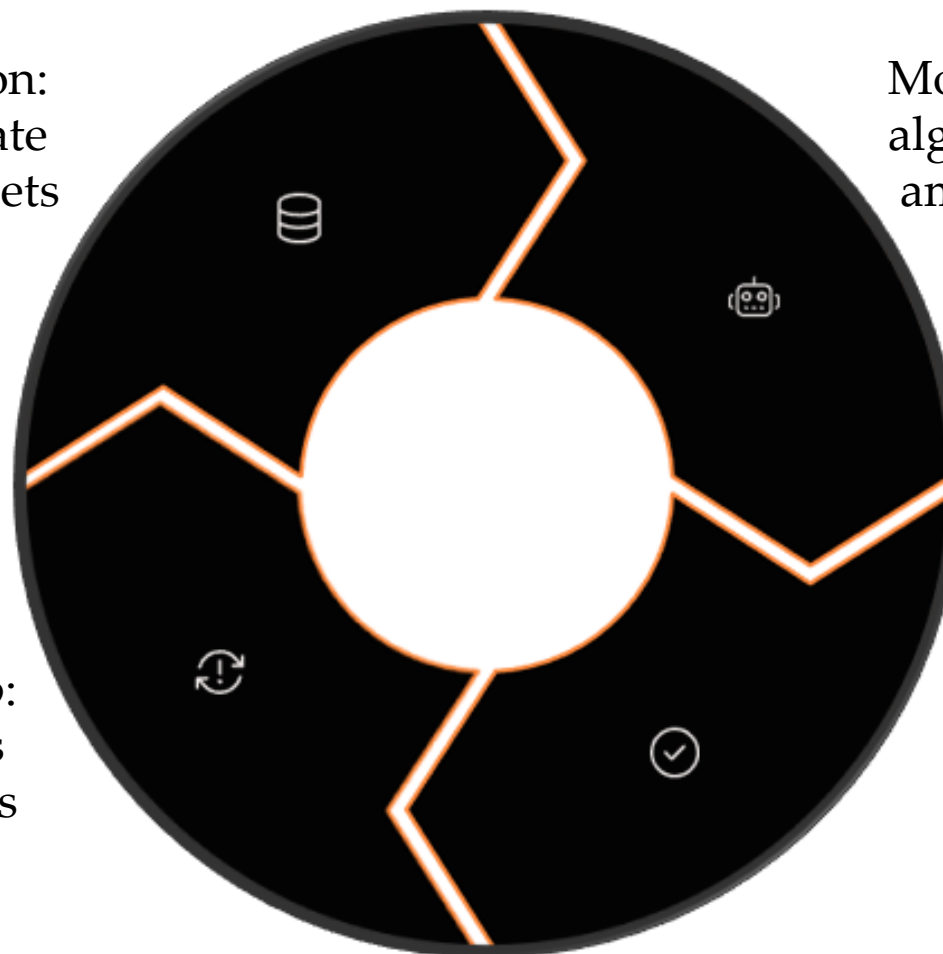
13.3 Bias Amplification

Biased data collection:
historical inequities create
imbalanced datasets

Model training:
algorithms learn and
amplify patterns of bias

Feedback loop:
new data reinforces
existing patterns

Deployment decisions:
biased predictions lead
to unfair outcomes



Why ML Bias Matters

13.2 Introduction to ML Bias

- ML bias can have a significant impact on society
 - Impact on discrimination: biased AI can lead to discriminatory decisions, affecting hiring, lending, policing, and healthcare
 - E.g., biased facial recognition algorithms may lead to wrongful arrests or surveillance of marginalized communities
 - E.g., predictive algorithms use historical crime data to predict future crimes: if past data overrepresents certain neighborhoods, ML models will produce biased outcomes
 - Impact on opportunities: AI bias can result in lost opportunities for demographic groups
 - E.g., biased hiring algorithms exclude qualified candidates based on gender, race, or socioeconomic background
- Importance of awareness of ML bias
 - For ML developers: being aware helps developers create fair ML systems, conduct bias testing, and promote the development of inclusive datasets for model training
 - For society: a better-informed public can hold AI developers and companies accountable



Case Study: COMPAS

13.4 Case Study: COMPAS

- COMPAS is an ML system that has been used in New York, Wisconsin, California, Florida, and other states
- COMPAS scores:
 1. Person commits a crime, is arrested
 2. COMPAS software predicts the chance that the person will commit another crime in the future (recidivism)
 3. Recidivism scores impact criminal sentences: if a person is likely to commit another crime, they may get a longer sentence
- [2016 ProPublica article](#) analyzed COMPAS scores for over 7,000 people arrested in Florida

Case Study: COMPAS

13.4 Case Study: COMPAS

- Error metrics

	Prediction: Low Risk	Prediction: High Risk
Outcome: No Recidivism	True Negative (TN)	False Positive (FP)
Outcome: Recidivated	False Negative (FN)	True Positive (TP)

- Error Rate** = $\frac{FP+FN}{TN+FP+FN+TP}$
 - How often is the prediction wrong (opposite to accuracy)
- False Positive Rate** = $\frac{FP}{FP+TN}$
 - False alarm:** How often were non-offenders predicted to reoffend?
 - A true negative (non-offender) is incorrectly classified as positive (offender)
- False Negative Rate** = $\frac{FN}{FN+TP}$
 - Miss:** How often were offenders predicted not to reoffend?
 - A true positive (offender) is incorrectly classified as negative (non-offender)

Case Study: COMPAS

13.4 Case Study: COMPAS

- Metrics by the COMPAS model

	Prediction: Low Risk	Prediction: High Risk
Outcome: No Recidivism	2681 (TN)	1282 (FP)
Outcome: Recidivated	1216 (FN)	2035 (TP)

- Error Rate = $\frac{FP+FN}{TN+FP+FN+TP} \approx 34.6\%$
- False Positive Rate = $\frac{FP}{FP+TN} \approx 32.4\%$
- False Negative Rate = $\frac{FN}{FN+TP} \approx 37.4\%$

Case Study: COMPAS

13.4 Case Study: COMPAS

- Metrics by the COMPAS model for Black defendants and White defendants

Black Defendants	Prediction: Low Risk	Prediction: High Risk
Outcome: No Recidivism	990 (TN)	805 (FP)
Outcome: Recidivated	532 (FN)	1369 (TP)

Error Rate $\approx 36.2\%$

False Positive Rate $\approx 44.9\%$

False Negative Rate $\approx 28.0\%$

White Defendants	Prediction: Low Risk	Prediction: High Risk
Outcome: No Recidivism	1139 (TN)	349 (FP)
Outcome: Recidivated	461 (FN)	505 (TP)

Error Rate $\approx 33.0\%$

False Positive Rate $\approx 23.5\%$

False Negative Rate $\approx 47.7\%$

- Similar error rates between white and Black defendants
- Black defendants have almost 2 times higher False Positive Rate (false alarm)
- White defendants have almost 2 times higher False Negative Rate (miss)



Case Study: COMPAS

13.4 Case Study: COMPAS

- COMPAS gives very different outcomes for White vs. Black defendants, but it does not use race as an input to the algorithm
- Even if a sensitive feature (e.g., race) is not an input to the algorithm, other features (e.g., zip code) may correlate with the sensitive feature
 - Correlated features that indirectly encode protected information are often called **proxy variables** or **proxy features**
 - Removing protected attributes alone does not guarantee fairness because the model can still learn discriminatory patterns from correlated variables
- This example illustrates why fairness evaluation must examine model outcomes and error rates across demographic groups, rather than examining only the model inputs

Formalizing Fairness

13.5 Fairness Metrics

- Fairness in ML can be defined in multiple mathematical ways, and multiple *fairness metrics* have been proposed and described on the next pages
 - Different definitions emphasize goals such as equal treatment, equal opportunity, or equal reliability of predictions
 - The fairness metrics may conflict with one another, making it impossible to satisfy all definitions simultaneously in many real-world applications
- Fairness in ML is not purely a technical problem, and also requires considering the application context, stakeholders, legal requirements, and societal impacts
 - The choice of fairness metric depends on the domain and on the consequences of different types of errors and decisions
- In practice, organizations must select the fairness criteria that are most appropriate for their use case and clearly communicate the resulting trade-offs

Statistical Parity

13.5 Fairness Metrics

- **Statistical Parity** (a.k.a. Demographic Parity): requires the probability of **receiving a positive outcome** to be equal across all groups
 - $P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b)$ for all values a, b of the protected attribute A
 - Notation
 - $\hat{Y}$ – prediction by the model
 - A – a protected attribute like race, gender, or age
 - a and b – two different groups
- E.g., in a loan approval system, the approval rate must be the same for all demographic groups, regardless of other factors
 - Let's assume the protected attribute A is gender, with groups $a = \text{female}$ and $b = \text{male}$
 - $\hat{Y} = 1$ means the model outputs a positive decision
 - $P(\hat{Y} = 1 \mid A = \text{female})$ is the probability the loan is approved for female applicants
 - If 100 female applicants applied, and 60 were approved, then: $P(\hat{Y} = 1 \mid A = \text{female}) = 0.6$
 - If 100 male applicants applied, and 80 were approved, then: $P(\hat{Y} = 1 \mid A = \text{male}) = 0.8$
 - $0.8 \neq 0.6$: statistical parity violated, fair model should have approximately the same approval rate for men and women

Equalized Odds, Equal Opportunity

13.5 Fairness Metrics

- **Equalized Odds:** requires that both **false positive** and **false negative rates** are equal across groups
 - E.g., the COMPAS example
 - False positive rate (false alarm): Black defendants (44.9%), White defendants (23.5%)
 - False negative rate (miss): Black defendants (28.0%), White defendants (47.7%)
- **Equal Opportunity:** focuses on fairness for **qualified individuals**, ensuring they have the same chance of favorable outcomes regardless of their protected characteristics
 - $P(\hat{Y} = 1 \mid Y = 1, A = a) = P(\hat{Y} = 1 \mid Y = 1, A = b)$
 - $\hat{Y} = 1$ means the model predicts a positive outcome, and $Y = 1$ means the applicant is qualified
 - Example
 - 100 qualified female applicants, 80 approved: $P(\hat{Y} = 1 \mid Y = 1, A = \text{female}) = 0.8$
 - 100 qualified male applicants, 60 approved: $P(\hat{Y} = 1 \mid Y = 1, A = \text{male}) = 0.6$
 - $0.8 \neq 0.6$: equal opportunity violated, creditworthy male applicants are less likely to be approved than equally creditworthy female applicants
 - Qualified individuals have equal chance of positive prediction regardless of group

Predictive Parity

13.5 Fairness Metrics

- **Predictive Parity:** focuses on fairness in **positive predictions**, ensuring they are equally reliable across all demographic groups
 - $P(Y = 1 \mid \hat{Y} = 1, A = a) = P(Y = 1 \mid \hat{Y} = 1, A = b)$
 - $\hat{Y}=1$ the model predicts a positive outcome, and $Y=1$ means the actual outcome is positive
 - When the model predicts a positive outcome, that prediction is equally likely to be correct for all demographic groups
 - E.g., approved loans have the same repayment rate ($Y = 1$) regardless of borrower demographics
 - Example
 - 100 female applicants approved, 80 repaid: $P(Y = 1 \mid \hat{Y} = 1, A = \text{female}) = 0.8$
 - 100 male applicants approved, 60 repaid: $P(Y = 1 \mid \hat{Y} = 1, A = \text{male}) = 0.6$
 - $0.8 \neq 0.6$: predictive parity violated
 - This metric is also called Positive Predictive Value (PPV) parity
- Equal opportunity vs predictive parity
 - Equal opportunity: if you deserve the opportunity (e.g., qualify for a loan), are you equally likely to receive it regardless of your group?
 - Predictive parity: If you're predicted to succeed (approved for a loan), is that prediction equally likely to be correct regardless of your group?

Calibration

13.5 Fairness Metrics

- **Calibration:** a model is calibrated if predicted **probability risk score** matches the observed outcome rate across all groups
 - $P(Y = 1 \mid S = s, A = a) = s$, for all groups and probability risk scores s
 - Notation:
 - S – the continuous probability risk score output by the model (value between 0 and 1)
 - s – a specific score value (e.g., $s = 0.7$ means 70% predicted risk)
 - E.g., among all applicants in a group with a predicted probability score by the model of 70%, approximately 70% should actually default
 - 100 female and 100 male loan applicants are both scored at 70% risk
 - 70 females actually defaulted: $P(Y = 1 \mid S = 0.7, A = \text{female}) = 0.7$, calibrated model
 - 50 males actually defaulted: $P(Y = 1 \mid S = 0.7, A = \text{male}) = 0.5 \neq 0.7$, not calibrated model
 - Calibration is especially important when the probability risk scores drive decisions (such as when a bank officer uses 70% scores as high-risk)
 - Calibration is related to predictive parity
 - Predictive parity checks accuracy at binary decisions (yes/no predictions), while calibration check accuracy for continuous risk scores (30% risk or 70% risk)

Individual and Intersectional Fairness

13.5 Fairness Metrics

- **Individual Fairness:** similar individuals should be treated similarly regardless of protected characteristics
 - $D(x_i, x_j)$ is a task-specific similarity metric between two individuals i and j , where similar individuals have a small distance
 - $|f(x_i) - f(x_j)|$ is the difference in model outputs between the two individuals
 - $|f(x_i) - f(x_j)| \leq L \cdot D(x_i, x_j)$
 - E.g., two applicants with similar income, credit history, and debt level should receive similar loan approval scores, regardless of gender
 - Challenge is that defining a fair similarity metric $D(x_i, x_j)$ is context-dependent
- **Intersectional Fairness:** considers combinations of attributes (e.g., Black women as a distinct group)
 - Standard group fairness evaluates one protected attribute at a time (such as race or gender)
 - A model can be fair with respect to race and fair with respect to gender, yet still discriminate against Black women
 - E.g., in a loan approval system, approval rates may be equal across gender and equal across race, yet Black women may still face lower approval rates than any other group



The Measurement Problem in Fairness

13.5 Fairness Metrics

- A fundamental challenge in calculating fairness metrics is the assumption that the ground-truth labels Y in the training data accurately reflect reality
 - In practice, ground-truth labels may themselves encode historical injustice
- Examples of biased labels
 - "Creditworthy" labels ($Y = 1$) defined by banks with decades of discriminatory lending history
 - "Qualified" hiring labels reflect unequal access to education and opportunity
- *The measurement problem*
 - Fairness depends on what we measure
 - Correcting for bias in predictions does not fix bias in the underlying labels, and all the fairness metrics we discussed depend on those labels being trustworthy
- Practitioners should critically examine whether training labels are valid and fair before optimizing for them
 - Consider whether the prediction target itself embeds historical inequality

Impossibility of Simultaneous Fairness

13.5 Fairness Metrics

- *Impossibility of Simultaneous Fairness* ([Chouldechova, 2017](#))
- This study states that many commonly used fairness criteria are mutually incompatible and cannot generally be satisfied simultaneously
 - E.g., if base rates (such as people who default on a loan) differ across groups $P(Y = 1|A = a) \neq P(Y = 1|A = b)$, then Equalized Odds and Calibration metrics cannot simultaneously hold
 - At least one fairness criterion must be violated when base rates differ across groups
- The impossibility results do not imply fairness cannot be improved, but rather, they show that different fairness goals may conflict and require trade-offs
- Implications for practitioners
 - Choosing a fairness metric is a value judgment, not a technical problem
 - There is no universally correct definition of fairness
 - Always explicitly state which definition is used, why, and what trade-offs are accepted
 - Communicate clearly to stakeholders which criteria are and are not being optimized

Fairness vs. Accuracy Trade-offs

13.5 Fairness Metrics

- Imposing fairness constraints typically reduces model accuracy
 - This is the basic **fairness-utility trade-off**
- Different types of errors carry different social costs depending on the application
 - False positive (wrongly deny loan): financial and opportunity harm to the individual
 - False negative (approve risky loan): financial loss to the lending institution
 - Fairness-aware design must account for asymmetric social costs of different error types
- Stakeholder-dependent definitions
 - Bank optimizes profit, minimizes false negatives
 - Regulator optimizes equity, minimizes disparate impact across groups
 - Individuals want accurate, individualized, non-discriminatory assessments
- Fairness is a design choice under stakeholder conflict, and not a pure constraint

Causal Fairness

13.5.1 Causal Fairness and Counterfactual Methods

- Statistical fairness metrics measure **correlations** in outcomes, but they do not capture underlying **causes**
 - For instance, a model can satisfy statistical parity and still be causally unfair
- Fairness decisions require **causal reasoning** about how features are generated and what they represent
 - E.g., the COMPAS model does not use race as an input feature to the model, however, it uses features such as zip code, prior arrests, and family history
 - These features act as **proxies** for race due to historical and structural discrimination
 - Removing race from input features does not eliminate its influence on predictions
- Example
 - Causal path: Gender -> Occupation -> Salary
 - If gender influences occupation (through discrimination), occupation becomes a proxy for gender
 - Statistical correction for occupation removes discrimination effects, but also removes job-related differences
 - To understand why the relationship exists requires a causal model

Structural Causal Models

13.5.1 Causal Fairness and Counterfactual Methods

- **Structural Causal Models** (SCMs) provide a formal language for causal fairness reasoning
 - A Directed Acyclic Graph (DAG) consists of nodes and edges, where nodes are variables/features, and edges are causal relationships
 - E.g., a protected attribute A (e.g., race) may influence feature X through historical mechanisms
 - A causally fair model should attempt to block discriminatory causal paths from A to Y that represent unjust discrimination
- Types of causal paths
 - Direct effect: $A \rightarrow Y$ (direct discrimination on the protected attribute)
 - Indirect effect via proxy: $A \rightarrow X \rightarrow Y$ (indirect discrimination through correlated features)
 - Acceptable: $A \rightarrow X \rightarrow Y$ only if X is a legitimate qualification independent of discrimination
- Example in criminal justice
 - Graph path: Race $\rightarrow$ Neighborhood $\rightarrow$ Arrest History $\rightarrow$ Risk Score
 - Arrest history is a proxy for race via biased policing patterns, not actual criminal behavior
 - A causally fair model should block information flowing through this discriminatory path
- Identifying causal paths requires domain knowledge and cannot be determined from data alone

Counterfactual Fairness

13.5.1 Causal Fairness and Counterfactual Methods

- *Counterfactual fairness* ([Kusner, 2017](#))
 - A decision is fair if it would be the same in a counterfactual world where the individual belonged to a different demographic group, with all else held equal
 - E.g., "Would this person have received the same outcome if born into a different group?"
- Relation to group fairness
 - Group fairness (statistical parity) compares aggregate outcomes between groups
 - Counterfactual fairness compares outcomes for the same individual across hypothetical worlds
 - Counterfactual fairness is more granular, as it evaluates fairness at the individual level
- Challenge is that counterfactual fairness requires a causal model (SCM)
 - The causal model must be specified and validated, which is non-trivial in practice
 - Different causal assumptions can lead to different fairness conclusions



Bias and Fairness Challenges with LLMs

13.6 Bias and Fairness in Large Language Models

- LLMs introduce new and complex bias and fairness challenges
- Reinforcement Learning from Human Feedback (RLHF) and alignment bias
 - RLHF aligns LLMs via human rater preferences, however, rater pools typically reflect demographic biases
 - Models aligned to what biased raters find "helpful" encode those biases as correct behavior
- Political bias
 - LLMs trained on internet data reflect dominant cultural and political perspectives
 - Studies show measurable political biases in major commercial LLMs
- Prompt sensitivity as a fairness issue
 - Small demographic hints in prompts significantly alter model outputs
 - E.g., identical resumes with stereotypically Black vs. White names receive different AI evaluations
 - Affects AI tools used for hiring, education, healthcare, content moderation at scale



Testing for Bias in Language Models

13.6 Bias and Fairness in Large Language Models

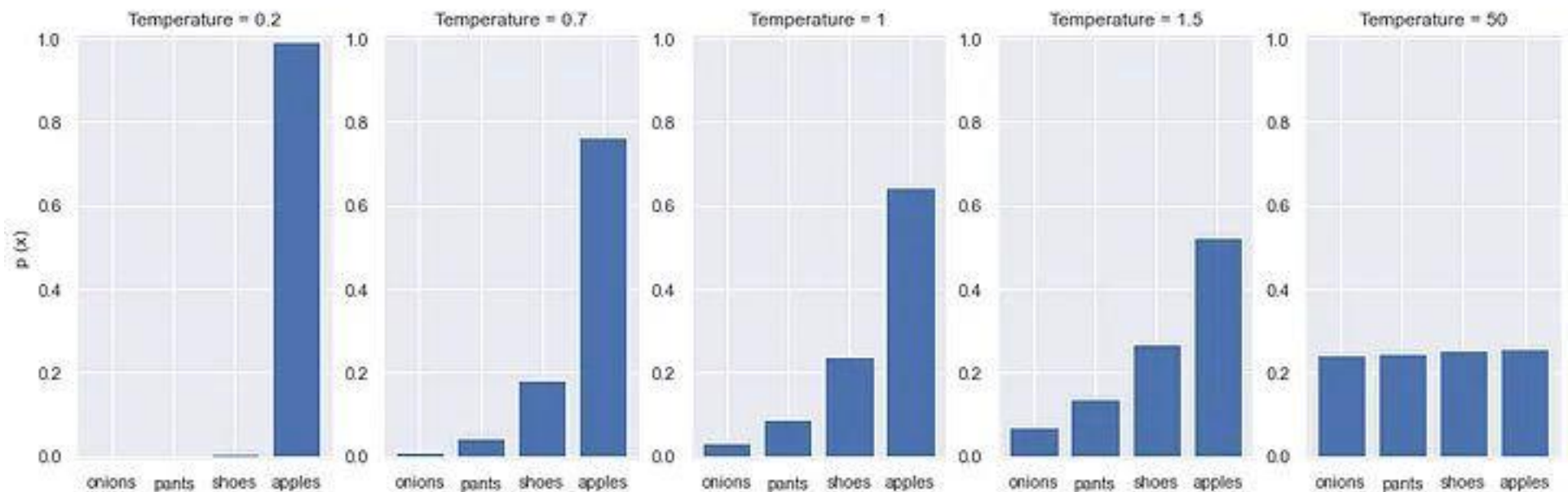
- [Kotek \(2023\) “Gender bias and stereotypes in Large Language Models”](#)
 - Researchers tested prompts like “The doctor phoned the nurse because she was late for the morning shift.”
 - The AI often inferred that the nurse was female and the doctor male
- Findings
 - On average, models were 6.8 times more likely to assign stereotypical female occupations to female pronouns and male occupations to male pronouns
 - Reinforces gender stereotypes, potentially affecting how language models are applied in industries like hiring and social media
- Impact
 - This kind of bias, if unchecked, may influence societal expectations and perpetuate harmful stereotypes
 - Biases affect users’ trust in AI and the decisions influenced by AI

Randomness and Bias in LLMs

13.6 Bias and Fairness in Large Language Models

- To mitigate the potential harms of biased training data, LLM developers often implement safeguards such as fine-tuning and alignment techniques to reduce biased outputs
 - However, increasing the model's temperature introduces more randomness into its responses, which can reduce the consistency and reliability of the safeguards
 - This raises concerns about the effectiveness of bias mitigation under high-temperature settings

"I like red ____"



Bias in Word Embeddings

13.6 Bias and Fairness in Large Language Models

- *Word embedding models* (Word2Vec, GloVe) encode societal biases present in training corpora
 - "man" is to "doctor" as "woman" is to "nurse"
 - "man" is to "computer programmer" as "woman" is to "homemaker"
 - These analogies reflect stereotypes in the embedding vector space
- Embeddings are foundational components of modern LLMs
 - Biases propagate to downstream tasks: search, recommendation, hiring, translation
- Debiasing approaches
 - Hard debiasing: project out the "gender direction" from the embedding space
 - Soft debiasing: regularize training to reduce stereotypical associations

Mitigation Strategies: Pre-processing

13.7 Mitigation strategies

- *Pre-processing techniques* focus on addressing bias in the training data
 - By modifying the dataset before ML model training, these approaches can help prevent biases from being encoded into the model
 - Pre-processing strategies are model-agnostic and can be applied regardless of the specific algorithm being used
- Pre-processing techniques include:
- **Data reweighing**
 - Assign different weights to training examples to counteract representation imbalances
 - Examples from underrepresented groups receive higher weights, increasing their influence during model training
 - Reduces impact of majority group dominance
 - Preserves all original data points
 - Helps model learn patterns from minority groups

Mitigation Strategies: Pre-processing

13.7 Mitigation strategies

- **Strategic resampling**
 - Modify the training dataset by either duplicating instances from underrepresented groups (over-sampling) or removing instances from overrepresented groups (under-sampling)
 - Creates more balanced class distribution
 - Relatively simple to implement
 - Can be combined with other techniques
- **Synthetic data generation**
 - Create artificial data points to increase representation of underrepresented groups
 - E.g., using techniques like SMOTE (Synthetic Minority Over-sampling Technique) or generative models
 - Addresses data imbalance
 - Creates novel training examples



Mitigation Strategies: In-processing

13.7 Mitigation strategies

- *In-processing techniques* modify the ML algorithm to incorporate fairness considerations during training
 - This can be achieved via constrained optimization methods that balance prediction accuracy with fairness metrics
- Advantages
 - Incorporate fairness constraints directly into the learning algorithm
 - Add regularization terms that penalize unfair solutions during training
 - Optimize for both accuracy and fairness as dual objectives

Mitigation Strategies: Post-processing

13.7 Mitigation strategies

- *Post-processing techniques* adjust the ML predictions after training is complete, such as applying different thresholds for different groups to match error rates
 - These approaches offer flexibility but may sacrifice some predictive performance to achieve fairness goals
- Techniques
 - Adjust model outputs after training to satisfy fairness criteria
 - Apply different decision thresholds for different groups
 - Calibrate probability estimates across protected groups
- Example: identify uncertain predictions near decision boundary and apply fairness-oriented rules
- While pre-processing strategies focus on the data, in-processing and post-processing strategies address bias in the model output

Adversarial Debiasing

13.7 Mitigation strategies

- Adversarial debiasing ([Zhang, 2018](#)) removes demographic information via adversarial training
- The proposed model architecture consists of:
 1. **Predictor network f** that takes input features X and predicts target outcome Y
 2. **Adversary network g** takes the output of the predictor network f and predicts a protected attribute A
- During training, the adversary g attempts to recover protected attribute A from predictor's output, and the predictor f is trained to prevent g from recovering A
 - As a result, the predictor f learns representations that are predictive of Y but not predictive of the protected attribute A
 - Demographic information is effectively removed from the learned intermediate representation
- The advantages is that this approach directly optimizes fairness during training
 - Limitations include adversarial training instability, does not guarantee all fairness criteria, computationally expensive

Fairness Checklist in ML

13.8 Governance, Legal Frameworks, and Ethical AI

- Step 1 - Problem definition
 - Identify who is affected by model decisions and potential harms
 - Determine which demographic groups are protected by law and ethical considerations
- Step 2 - Data audit
 - Analyze training data for representation imbalance of demographic subgroups
 - Identify potential features for protected attributes
- Step 3 - Metric selection
 - Choose appropriate fairness metrics based on context
 - State which fairness criteria are not being optimized
- Step 4 - Evaluation by subgroup
 - Evaluate all performance metrics by protected attribute and intersections
 - Include fairness metrics in addition to accuracy metrics in all evaluation reports
- Step 5 - Mitigation
 - Apply pre/in/post-processing techniques based on context and constraints
 - Evaluate the fairness-accuracy trade-off
- Step 6 - Deployment and monitoring
 - Implement ongoing fairness monitoring in production systems



Governance and Legal Frameworks

13.8 Governance, Legal Frameworks, and Ethical AI

- Fairness assessments are becoming mandatory and required by law
- *EU AI Act (2024)*
 - First comprehensive AI regulation globally
 - Classifies AI systems into four risk tiers: unacceptable, high, limited, and minimal risk
 - High-risk AI (e.g., for hiring, credit, education, law enforcement, critical infrastructure) must meet strict bias and fairness requirements:
 - Conduct bias and fairness impact assessments before deployment
 - Maintain detailed technical documentation and audit logs
 - Provide human oversight and intervention mechanisms for consequential decisions
- *U.S. guidance documents*
 - Consumer Financial Protection Bureau (CFPB) guidance: prohibits discriminatory use of AI in credit decisions
 - Equal Employment Opportunity Commission (EEOC) guidance: AI-driven hiring tools must comply with employment discrimination law and other anti-discrimination laws
 - Executive Order on AI: directs federal agencies to evaluate and mitigate AI bias



Towards Ethical AI

13.8 Governance, Legal Frameworks, and Ethical AI

- Building fair and unbiased ML systems requires continuous effort
 - It requires vigilance at every stage of the AI lifecycle, from problem formulation to deployment and monitoring
- Transparency by ML developers is crucial, both in documenting design choices and in communicating limitations to users and stakeholders
 - This includes acknowledging trade-offs between different fairness criteria and being honest about the imperfect nature of all fairness solutions



Steps Towards Ethical AI

13.8 Governance, Legal Frameworks, and Ethical AI

- Education and awareness
 - Building fairness literacy among practitioners
- Community engagement
 - Including affected communities in design and evaluation
- Transparency and explainability
 - Making AI systems understandable to stakeholders
- Multidisciplinary collaboration
 - Bringing together technical, ethical, legal, and domain experts
- Continuous improvement
 - Ongoing monitoring, auditing, and refinement of ML systems

Bias and Unfairness in Information Retrieval Systems: New Challenges in the LLM Era

Kyle Russel Delos Santos
CS5727 Adversarial Machine Learning
05/05/2025



Introduction & Integration Lifecycle

- Paradigm Shift
- Core Problem: Bias and Unfairness
- Societal Risks: contribute to echo chambers



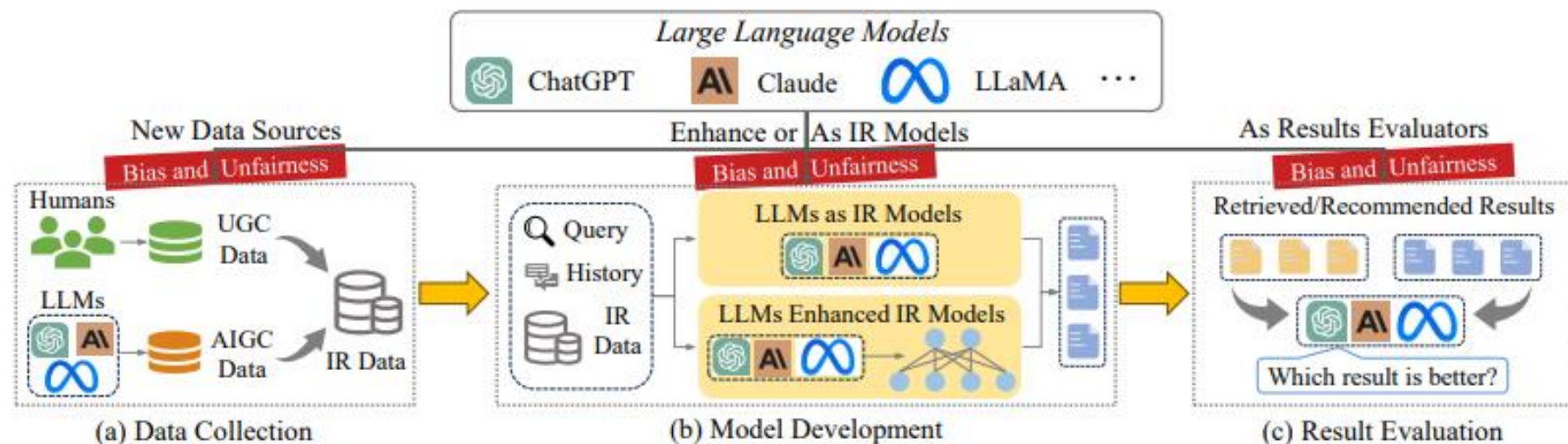


Figure 1: Overview of three stages of the intersection between LLMs and IR systems. (a) LLMs-generated content as new data sources for IR. (b) Incorporating LLMs to enhance or as IR models. (c) Adopting LLMs as results evaluators in IR systems.

Bias Across the Pipeline

- Data collection:
 - Source bias
 - Factuality bias
- Model Development:
 - Position bias
 - Popularity bias
 - hallucination-related bias
- Result Evaluation:
 - Selection bias:
 - Style bias
 - Egocentric bias



Unified View from Distribution Alignment

- Mathematical Framework:
 $P(R) = P(R)$
- Bias: mismatch between
factual & objective realities
- Unfairness: require alignment
w/ human concepts of equity
and distributive justice
- Gap



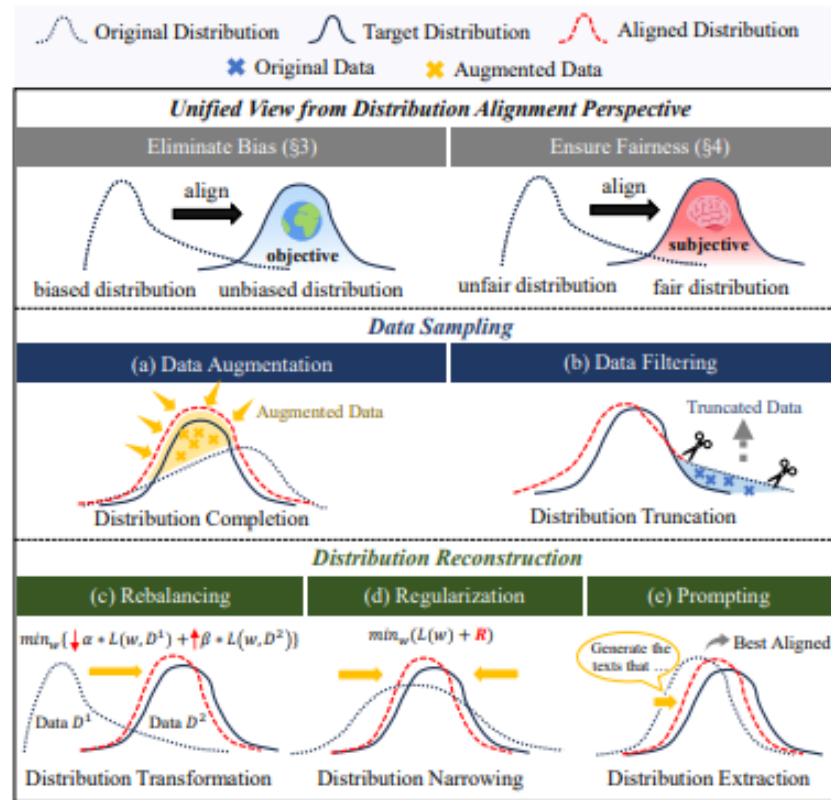


Figure 2: Illustration of different types of mitigation strategies from a unified view of distribution alignment.

Unfairness Breakdown

- User unfairness
- Item unfairness:
 - Exposure bias
- Evaluation unfairness



Taxonomy of Bias

Table 1: The taxonomy of different types of bias in the intersection between LLMs and IR systems.

Sourced Stage	Type	Mitigation Strategies				
		Data Sampling		Distribution Reconstruction		
		Data Augmentation	Data Filtering	Rebalancing	Regularization	Prompting
Data Collection	Source Bias		[18]		[28, 174, 200]	
	Factuality Bias	[51, 119, 126, 175–177, 184]	[51, 147, 182]			[119, 143, 159, 176]
Model Development	Position Bias	[58, 96, 123, 146, 166, 191]		[97, 166]		[58]
	Popularity Bias	[158, 191]				[31, 58, 140]
	Instruction-Hallucination Bias	[106, 131, 160]			[39]	[117, 183]
	Context-Hallucination Bias	[7, 42]				
Result Evaluation	Selection Bias	[21, 23, 79, 85, 116, 155, 196, 198]		[94, 155, 195]		[70, 116, 155, 196]
	Style Bias					[168, 196]
	Egocentric Bias	[79]		[91]		[56, 91]



Behavioral & Evaluation Biases

- Position Bias: LLM-based IR models favor documents at the beginning or end of a list
- Selection Bias: LLM evaluators are sensitive to order or ID tokens, like favoring choice 'C' or 'A'
- Style and Egocentric Bias: prefer longer, polished responses or outputs generated by themselves (egocentric)

Understanding Unfairness

Bias and Unfairness in Information Retrieval Systems: New Challenges in the LLM Era

KDD '24, August 25–29, 2024, Barcelona, Spain

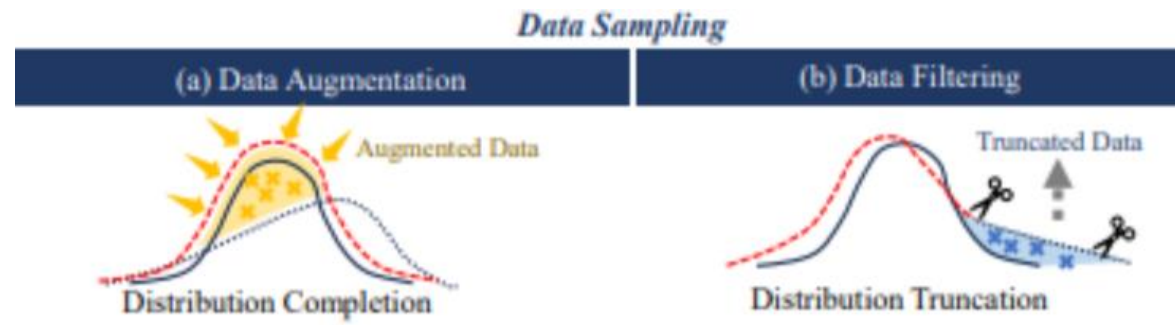
Table 2: The taxonomy of different types of unfairness in the intersection between LLMs and IR systems.

Sourced Stage	Type	Mitigation Strategies				
		Data Sampling		Distribution Reconstruction		
		Data Augmentation	Data Filtering	Rebalancing	Regularization	Prompting
Data Collection	User Unfairness	[47, 95, 141, 150, 170, 190]	[108, 125]	[32, 111]	[12, 62, 121]	[38]
	Item Unfairness	[127, 204]	[50]	[64]		[38, 73]
Model Development	User Unfairness	[152]	[102, 133, 137, 152]	[54, 187]	[6, 46, 89, 112, 114, 156, 164, 199]	[32, 59, 180, 190]
	Item Unfairness	[205]	[25, 69]	[64]	[40]	[31, 82, 205]
Result Evaluation	User Unfairness	[67]	[81]			[8, 63, 113, 128, 181]
	Item Unfairness	[49]		[5, 135]		[130, 151, 154, 189, 191]



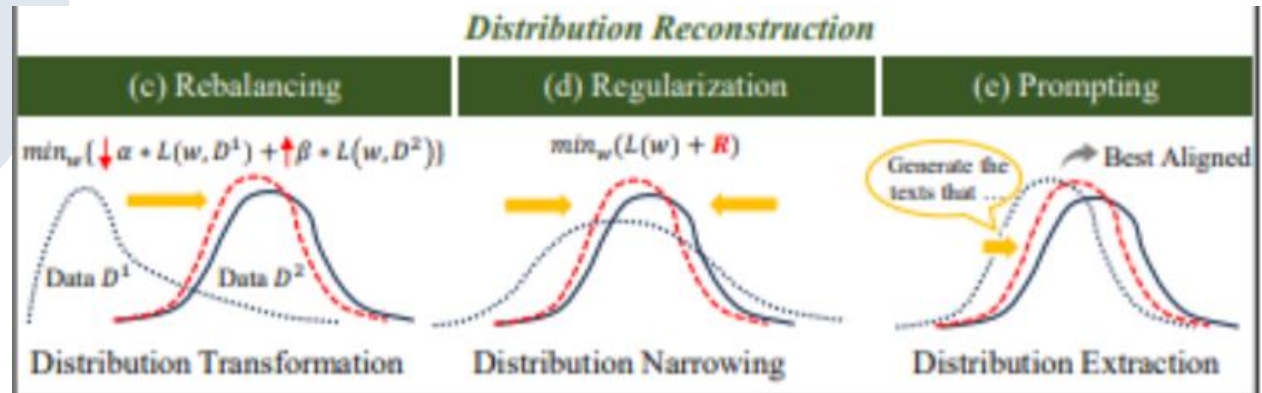
Mitigation Strategies

- Data augmentation
- Data filtering



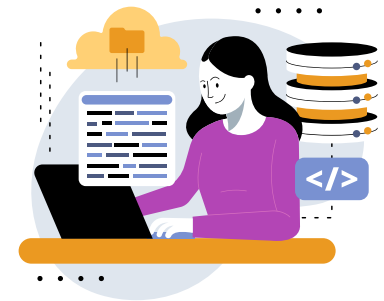
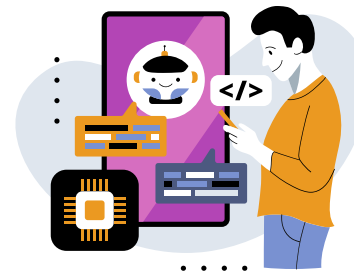
Distribution Reconstruction

- Rebalancing
- Regularization
- Prompting



Conclusion & Future Work

- Unified Solutions: Moving beyond addressing individual instances of bias toward integrated, unified mitigation frameworks
- Interrupting Feedback Loops: Stop the cycle where biased information influences future training data
- Dynamic Benchmarks: Transitioning from simulated environments to large-scale, real world. and dynamic evaluation protocols.





References

- Aditi Rajesh Shah, Nick Szydłowski, “AI Bias: A Hands-On Workshop and Discussion,” available at <https://tiny.sjsu.edu/ai-bias>
- Justin Joshnson, David Fouhey, “Lecture: Fairness in AI,” EECS 442, University of Michigan, 2021